



SIGNALS PROJECT Report

This project divides to 4 Sections, the output in every section explained with photos and comments, hope you find it good enough for full mark ^^.

By Ahmed Sami Zubaidia.

Student Number: 1200105

Section One: -

First as a result of the ID number

The variables A ,B ,and C it will be 1,5,5 Respectively

Qusestion 1 :-

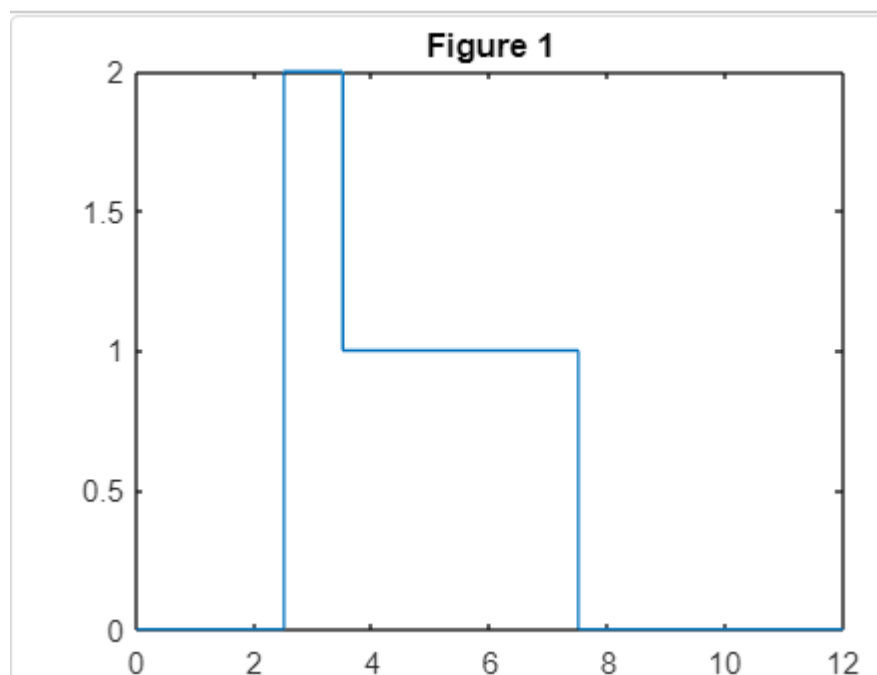
1. Generate and Plot the following signals

$$A. \ x(t) = \Pi[(t-3)/A] + \Pi[(t-C)/B]$$

Answer: -

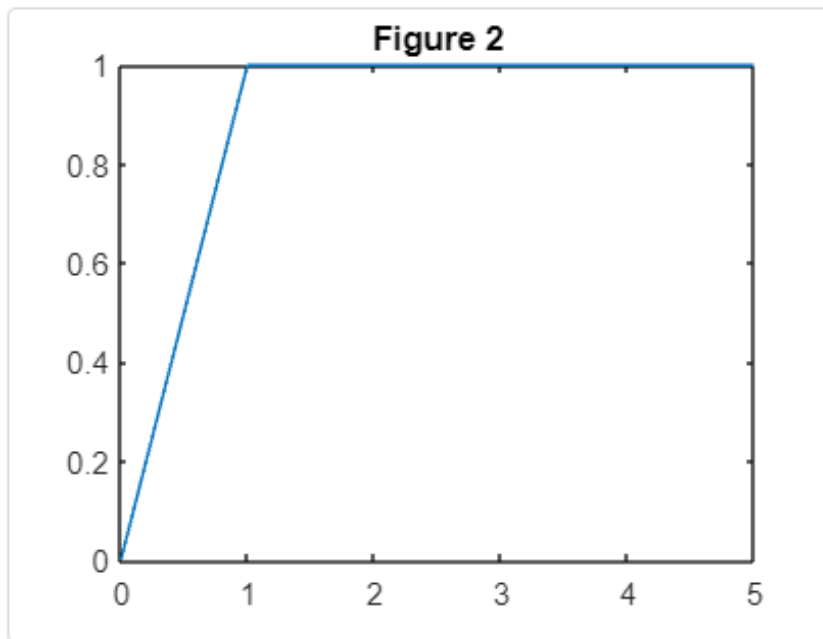
Q1) A

```
syms t;  
x1 =rectangularPulse((1/1)*(t-3)) +rectangularPulse((1/5)*(t-5));  
fplot(x1,[0,12])  
title("X1")
```



$$B. \quad x_b(t) = r(t) - r(t-A) - r(t-B) + r(t-C)$$

```
syms t;
r = @(t) heaviside(t) * t;
x2 = @(t) r(t) - r(t-1) - r(t-5) + r(t-5);
fplot(x2(t),[0,5])
title("Figure 2")
```



Comment Section Question 1: -

Syms: in order to declare a variable, we will use it as an input in our functions.

@t: A function handle is a MATLAB® data type that stores an association to a function. Indirectly calling a function enables you to invoke the function regardless of where you call it from.

fplot: fplot(f) plots the curve defined by the function $y = f(x)$ over the default interval $[-5 \ 5]$ for x .

heaviside function: unit step function.

Section Two: -

As a result of the ID number

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Qusestion 2 :-

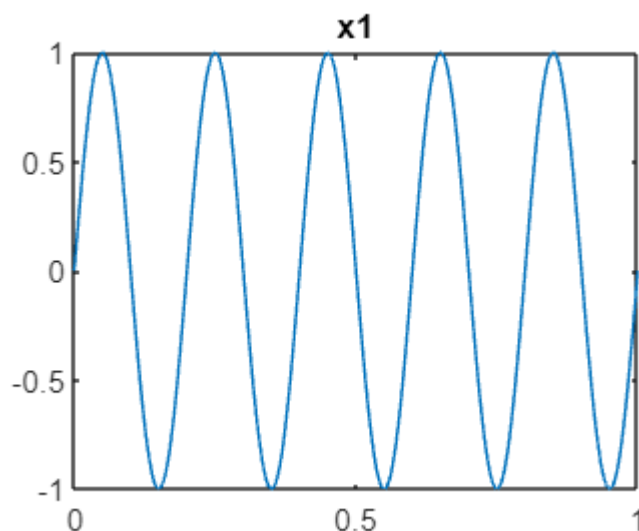
2. Consider the following signals:

$$x_1(t) = A \sin(10\pi t), \quad x_2(t) = \frac{1}{3}A \sin(30\pi t), \quad x_3(t) = \frac{1}{5}A \sin(50\pi t)$$

A. Generate and plot $x_1(t)$ for one period.

```
syms t;  
x1 = @(t) 1*sin(10 * pi * t);
```

```
fplot(x1,[0,1])  
title("x1")
```



Periodic As the plot showing.

2. Consider the following signals:

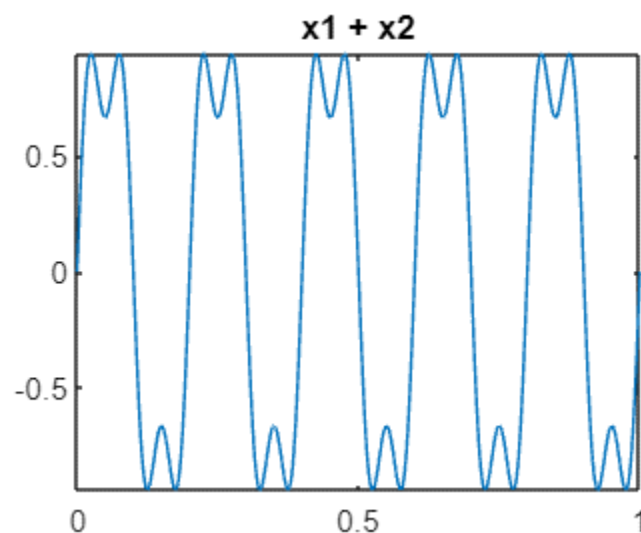
$$x_1(t) = A \sin(10\pi t), \quad x_2(t) = \frac{1}{3}A \sin(30\pi t), \quad x_3(t) = \frac{1}{5}A \sin(50\pi t)$$

B. Generate and plot $x_b(t) = x_1(t) + x_2(t)$ for one period.

```
|syms t;  
x1 = @(t) 1*sin(10 * pi * t);  
x2 = @(t) (1/3)*1*sin(30 * pi * t);  
x3 = @(t) (1/5)*1*sin(50 * pi * t);
```

```
m = @(t) x1(t) + x2(t);
```

```
fplot(m,[0,1])  
title("x1 + x2")
```



Periodic As the plot showing.

2. Consider the following signals:

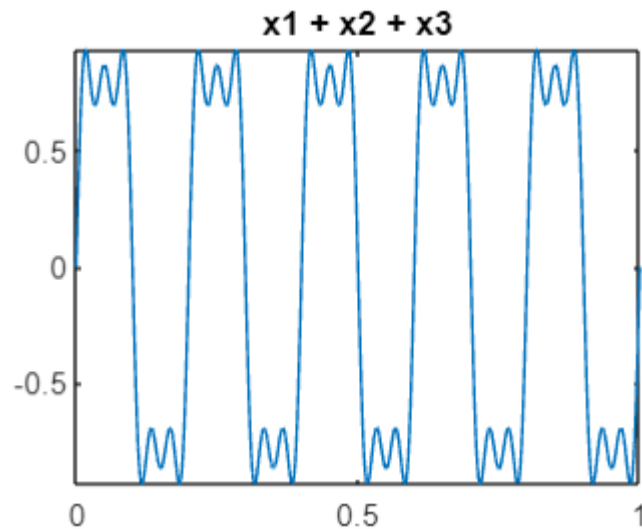
$$x_1(t) = A \sin(10\pi t), \quad x_2(t) = \frac{1}{3}A \sin(30\pi t), \quad x_3(t) = \frac{1}{5}A \sin(50\pi t)$$

C. Generate and plot $x_c(t) = x_1(t) + x_2(t) + x_3(t)$ for one period.

```
syms t;  
x1 = @(t) 1*sin(10 * pi * t);  
x2 = @(t) (1/3)*1*sin(30 * pi * t);  
x3 = @(t) (1/5)*1*sin(50 * pi * t);
```

```
n = @(t) x1(t) + x2(t) + x3(t);
```

```
fplot(n,[0,1])  
title("x1 + x2 + x3")
```



Periodic As the plot showing.

Section Three: -

As a result of the ID number

The variables A ,B ,and C it will be 1,5,5 Respectively

Qusestion 3 :

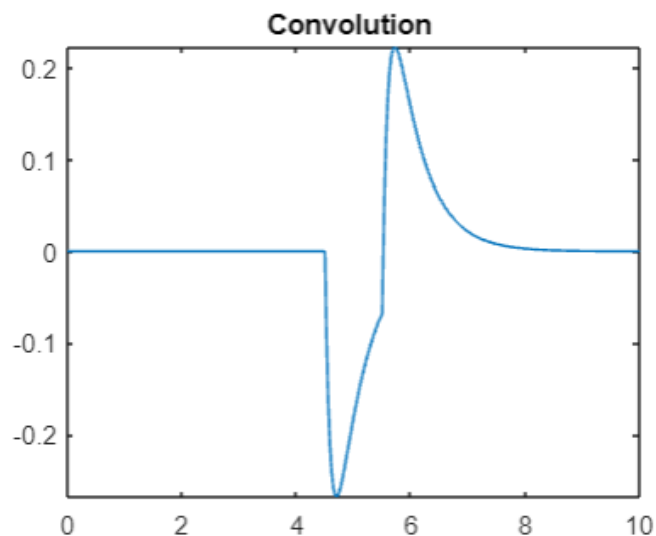
3. Find and sketch the signal $y(t)$ which is the convolution of the two pairs if signals.

$$x(t) = [e^{-2t} - Ce^{-10t}]u(t), \quad h(t) = \Pi\left(\frac{t-B}{A}\right)$$

```
syms t tow;
x = (exp(-2*tow)-5*exp(-10*tow)).*heaviside(tow);
h = rectangularPulse((1/1)*(t-tow -5));
y= int(x*h,tow,-inf,inf);

fplot(y,[0,10])
title("Convolution ")

disp("Solution: ")
disp(y)
```



Solution:

$$\frac{\text{heaviside}\left(t - \frac{11}{2}\right) e^{-10t} e^{11} (e^{8t} - e^{44})}{2} - \frac{\text{heaviside}\left(t - \frac{9}{2}\right) e^{-10t} e^9 (e^{8t} - e^{36})}{2}$$

Section Forth: -

As a result of the ID number

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Qusestion 4

4. For LTI system $h(t) = Ae^{-Bt}$, consider the input square wave be:

$$x(t) = A + \sum_{\substack{k=-\infty \\ k \text{ odd} \\ k \text{ odd}}}^{\infty} \frac{B}{\pi k} e^{-j\pi/2} e^{jkt}$$

A. Plot the system frequency response (Amplitude and Phase).

```
%*****evaluate the summation*****%
sum =0;
t=-40 :0.1:40;

for n= -25 :1:25 % Summation from -25 to 25

    if(mod(n,2)==1) % cheack the if n odd

        Xt= (5/pi.*n).*exp((-1i.*pi)/2).*exp(1i*n.*t);
        sum = sum +Xt ;
    end

end % for n loop

sum =sum +1 ; %A =1

%***** End *****%

%***** frequency response evaluate *****%
```

```

Xt=sum;
h = exp(-5.*t);

Xf= fft(Xt); %fourier transform xt
hf= fft(h); %fourier transform h(t)

Y_Mag = abs(hf); % Magnitude of frequency response
Y_phase = angle(hf); % Phase of frequency response

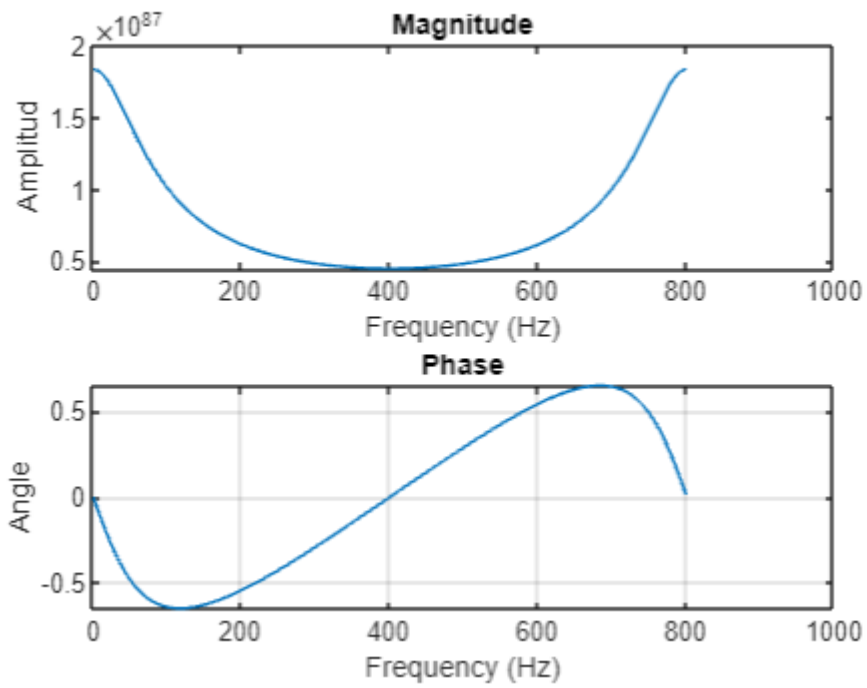
%***** frequency response evaluate End *****%

%ploting

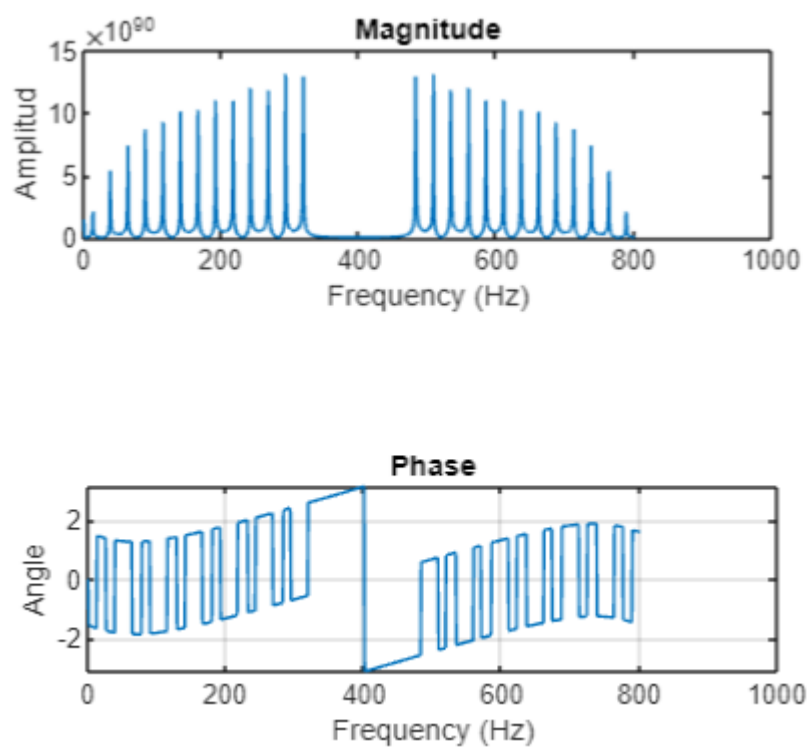
subplot(2,1,1);
plot(Y_Mag);
xlabel('Frequency (Hz)'),ylabel('Amplitud'),title('Magnitude');

subplot(2,1,2);
plot(Y_phase);
xlabel('Frequency (Hz)'),ylabel('Angle'),title('Phase');
grid on

```



$H(F)$



$Y(F)$

Y(F) code

```

yf= Xf.*hf;

Y_Mag2 = abs(yf); % Magnitude of frequency response
Y_phase2 = angle(yf); % Phase of frequency response

subplot(2,1,1);
plot(Y_Mag2);
xlabel('Frequency (Hz)'),ylabel('Amplitud'),title('Magnitude');

subplot(2,1,2);
plot(Y_phase2);
xlabel('Frequency (Hz)'),ylabel('Angle'),title('Phase');
grid on

```

4. For LTI system $h(t) = Ae^{-Bt}$, consider the input square wave be:

$$x(t) = A + \sum_{\substack{k=-\infty \\ k \text{ odd}}}^{\infty} \frac{B}{\pi k} e^{-j\pi/2} e^{jkt}$$

B. Plot the system time response for the square wave input
(consider the time interval [0:0.1:7])

```

%*****evaluate the summation*****%
sum =0;
t=0 :0.1:7;

for n= -25 :1:25 % Summation from -25 to 25

    if(mod(n,2)==1) % cheack the if n odd

        Xt= (5/pi.*n).*exp((-1i.*pi)/2).*exp(1i*n.*t);
        sum = sum +Xt ;
    end

```

```

end % for n loop

sum =sum +1 ; %A =1

%***** End *****%

%***** frequency response evaluate *****%

Xt=sum;
h = exp(-5.*t);

Xf= fft(Xt); %fourier transform xt
hf= fft(h); %fourier transform h(t)

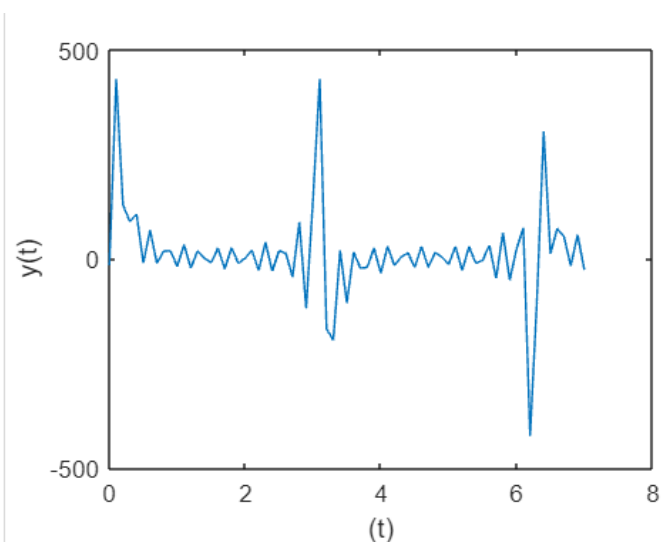
yf= Xf.*hf;
Y_Mag = abs(yf); % Magnitude of frequency response
Y_phase = angle(yf); % Phase of frequency response

%***** frequency response evaluate End *****%
yt = ifft( yf );

plot(yt,t);

xlabel('(t)'),ylabel('y(t)');

```



Y(t)